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Homework-6 (Relativistic quantum mechanics)

1. **Klein-Gordon equation:** The Klein-Gordon (KG) equation is given by

$$\frac{1}{c^2} \frac{\partial^2 \psi(\mathbf{r}, t)}{\partial t^2} = \nabla^2 \psi(\mathbf{r}, t) - \left(\frac{m_e c}{\hbar} \right)^2 \psi(\mathbf{r}, t),$$

where $\psi(\mathbf{r}, t)$ is a single-component wave function.

- (a) Obtain the following continuity equation from the KG equation:

$$\frac{\partial P}{\partial t} + \nabla \cdot \mathbf{J} = 0,$$

where the probability density is given by

$$P(\mathbf{r}, t) = \frac{i\hbar}{2m_e c^2} \left[\psi^*(\mathbf{r}, t) \frac{\partial \psi(\mathbf{r}, t)}{\partial t} - \psi(\mathbf{r}, t) \frac{\partial \psi^*(\mathbf{r}, t)}{\partial t} \right]$$

and the probability current density is given by

$$\mathbf{J}(\mathbf{r}, t) = \frac{\hbar}{2im_e} [\psi^*(\mathbf{r}, t) \nabla \psi(\mathbf{r}, t) - \psi(\mathbf{r}, t) \nabla \psi^*(\mathbf{r}, t)].$$

- (b) Show that the probability density is negative for negative energy branch.
What will be the probability density if $\psi(\mathbf{r}, t)$ is real?
- (c) Can we obtain/define a velocity operator from the KG equation?

2. **Properties of the Dirac matrices:** The Dirac matrices are given as

$$\alpha_j = \begin{pmatrix} 0 & \sigma_j \\ \sigma_j & 0 \end{pmatrix}, \beta = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, \quad (1)$$

where $j = x, y, z$, Pauli matrices σ_j , 2×2 null matrix 0 and 2×2 identity matrix I .

- (a) Check that $\alpha_j^2 = I_{4 \times 4}$ and $\beta^2 = I_{4 \times 4}$.
- (b) Show that $\{\alpha_i, \alpha_j\} = 2I_{4 \times 4} \delta_{ij}$ and $\{\alpha_i, \beta\} = 0$.
- (c) Show that

$$\alpha_j \alpha_k = i \epsilon_{jkl} \begin{pmatrix} \sigma_l & 0 \\ 0 & \sigma_l \end{pmatrix},$$

where $j, k, l : x, y, z$.

3. The Dirac equation for a free particle is given by

$$i\hbar \frac{\partial \psi(\mathbf{r}, t)}{\partial t} = H \psi(\mathbf{r}, t),$$

where the Dirac Hamiltonian H is given by

$$H = c \boldsymbol{\alpha} \cdot \mathbf{p} + \beta m_e c^2$$

and the Dirac spinor is represented as $\psi^\dagger = (\psi_1^*, \psi_2^*, \psi_3^*, \psi_4^*)$.

- (a) Write down the Dirac equation explicitly in 4×4 matrix form.
- (b) Show that the velocity operator for a Dirac particle is $\hat{\mathbf{v}} = c\boldsymbol{\alpha}$.
Note that the velocity operator in Dirac theory is a purely matrix operator $\boldsymbol{\alpha}$, as opposed to the differential operator (in real space) in non-relativistic case.
- (c) Obtain the following continuity equation from the Dirac equation:

$$\frac{\partial P}{\partial t} + \boldsymbol{\nabla} \cdot \mathbf{J} = 0,$$

where P is the probability density and $\mathbf{J}$ is the probability current density. Show that the expressions of P and $\mathbf{J}$ are $P(\mathbf{r}, t) = \psi^\dagger(\mathbf{r}, t)\psi(\mathbf{r}, t)$ and $\mathbf{J}(\mathbf{r}, t) = c\psi^\dagger(\mathbf{r}, t)\boldsymbol{\alpha}\psi(\mathbf{r}, t)$. Here $\psi(\mathbf{r}, t)$ is the four-component Dirac spinor.

4. Total angular momentum is a constant of motion:

- (a) Show that the orbital angular momentum $\mathbf{L} = \mathbf{r} \times \mathbf{p}$ is not a conserved quantity: $[\mathbf{L}, H] = i\hbar c(\boldsymbol{\alpha} \times \mathbf{p})$.
Orbital angular momentum is not a good quantum number for the Dirac equation.
- (b) Defining $\mathbf{S} = (\hbar/2)\boldsymbol{\Sigma}$, where components of the operator $\boldsymbol{\Sigma}$ are $\Sigma_x = -i\alpha_y\alpha_z$, $\Sigma_y = -i\alpha_z\alpha_x$ and $\Sigma_z = -i\alpha_x\alpha_y$. Show that

$$\boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\sigma} & 0 \\ 0 & \boldsymbol{\sigma} \end{pmatrix}.$$

- (c) Find the eigenvalues of $\mathbf{S}$. Show that $\mathbf{S}^2 = \frac{3\hbar^2}{4}I_{4 \times 4}$.
So it proves that $S = \hbar/2$ and the Dirac equation describes particles with spin 1/2. Thus $\mathbf{S}$ is the spin angular momentum operator for spin 1/2 particle.
- (d) Show that the total angular momentum is $\mathbf{J} = \mathbf{L} + \mathbf{S}$ is a constant of motion: $[\mathbf{J}, H] = 0$.
Total angular momentum is a good quantum number for the Dirac equation.
- (e) Show that the operator $\mathbf{S} \cdot \mathbf{p}$ is a constant of motion: $[H, \mathbf{S} \cdot \mathbf{p}] = 0$.
- (f) Helicity is defined as the projection of the spin along the motion of the particle. Mathematically, the helicity operator is defined as

$$\hat{h} = \frac{\boldsymbol{\Sigma} \cdot \hat{\mathbf{p}}}{|\mathbf{p}|}.$$

Find the eigenvalues and the eigenfunctions of the helicity operator $\hat{h}$.

Helicity is also a good quantum number.

5. Dirac spinors:

- (a) Write down the time-dependent Dirac equation. Obtain the time-independent Dirac equation in terms of the momentum $\mathbf{p}$ and the spinor $\psi = \begin{pmatrix} v \\ w \end{pmatrix}$, where $v = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ and $w = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}$.

- (b) Obtain the relativistic energy-momentum relation $E(\mathbf{p})$. Sketch $E(\mathbf{p})$ (in units of $m_e c^2$) vs the dimensionless momentum $|\tilde{\mathbf{p}}|$.
- (c) Obtain the four simplest spinors of the free Dirac particle with momentum $\mathbf{p}$. Identify spinors correspond to the positive as well as negative energy solutions.
- (d) Obtain the normalization constant of all the spinors correspond to both the positive and negative energy solutions. Calculate the average value of the velocity operator $\mathbf{v}$ and the spin angular momentum operator $\mathbf{S}$.

6. Non-relativistic approximation to the Dirac equation:

- (a) The Dirac equation for an electron in a hydrogen atom is given by

$$[c\boldsymbol{\alpha} \cdot \mathbf{p} + \beta m_e c^2 + V(r)]\Psi(\mathbf{r}) = E\Psi(\mathbf{r}),$$

where E is the total energy, $V(r)$ is the Coulomb interaction between nucleus and the electron and $\Psi(\mathbf{r}) = (\psi(\mathbf{r}), \chi(\mathbf{r}))$ with $\psi(\mathbf{r})$, and $\chi(\mathbf{r})$ are being two-component spinors. Show that in the non-relativistic limit ($p \ll m_e c$), it is reduced to the Schrödinger equation for the hydrogen atom along with the three relativistic correction terms (corrections to the kinetic energy, Darwin term and the spin-orbit interaction $\sim \mathbf{L} \cdot \mathbf{S}$ term).

- (b) Consider an electron described by the Dirac equation is subjected to the magnetic field $\mathbf{B}$. The Dirac equation is given by

$$[c\boldsymbol{\alpha} \cdot (\mathbf{p} + e\mathbf{A}) + \beta m_e c^2]\Psi(\mathbf{r}) = E\Psi(\mathbf{r}),$$

where $\mathbf{A}$ is a vector potential corresponding to the magnetic field $\mathbf{B}$ and E is the total energy. Here $\Psi(\mathbf{r}) = (\psi(\mathbf{r}), \chi(\mathbf{r}))$ with $\psi(\mathbf{r})$, and $\chi(\mathbf{r})$ are being two-component spinors. Show that in the non-relativistic limit, it reduces to

$$\left[\frac{(\mathbf{p} + e\mathbf{A})^2}{2m_e} + g \frac{e}{2m_e} \mathbf{S} \cdot \mathbf{B} \right] \psi(\mathbf{r}) = \epsilon \psi(\mathbf{r}),$$

where $g = 2$, $\mathbf{S} = (\hbar/2)\boldsymbol{\sigma}$ and $\epsilon = E - m_e c^2$ is the energy measured with respect to the rest mass energy.

It clearly demonstrate how the Zeeman interaction energy appears along with the correct g -factor and the spin angular momentum operator $\mathbf{S} = (\hbar/2)\boldsymbol{\sigma}$.